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Inventor(s)	Alissa; Husam Atallah et al.

Immersion cooling system that enables increased heat flux at heat-generating components of computing devices

Abstract

An immersion cooling system includes an immersion tank that is configured to retain dielectric working fluid and to hold a plurality of computing devices submerged in the dielectric working fluid. The immersion cooling system also includes a condenser that is configured to cause condensation of vaporized working fluid. The immersion cooling system also includes a subcooling heat exchanger that is in fluid communication with a coolant source. The coolant source provides coolant having a coolant temperature that is lower than a boiling point of the dielectric working fluid. The subcooling heat exchanger is positioned so that heat transfer can occur between the dielectric working fluid and the subcooling heat exchanger. The immersion cooling system also includes a control system that controls how much of the coolant flows into the subcooling heat exchanger based at least in part on a temperature of the dielectric working fluid.

Inventors: Alissa; Husam Atallah (Redmond, WA), Ramakrishnan; Bharath (Bellevue, WA), Manousakis; Ioannis (Redmond, WA), Keehn; Nicholas Andrew (Kirkland, WA), Peterson; Eric Clarence (Woodinville, WA)

Applicant: Microsoft Technology Licensing, LLC (Redmond, WA)

Family ID: 81392782

Assignee: Microsoft Technology Licensing, LLC (Redmond, WA)

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Primary Examiner: Nouketcha; Lionel

Attorney, Agent or Firm: Ray Quinney & Nebeker P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/212,961, filed Jun. 22, 2023, which claims priority to and the benefit of U.S. patent application Ser. No. 17/320,063, filed on May 13, 2021, which are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Computing devices include heat-generating components that produce significant amounts of heat during normal operation. Examples of such heat-generating components include central processing units (CPUs), graphics processing units (GPUs), tensor processing units (TPUs), memory devices, and other integrated circuits.

(2) Computer cooling is the process of removing heat generated by heat-generating components within a computing device to keep components within permissible operating temperature limits. Cooling can be important because computer components are susceptible to temporary malfunction or permanent failure if they are overheated.

(3) A datacenter is a physical facility that is used to house computing devices and associated components. A datacenter typically includes a large number of computing devices (e.g., servers), which may be stacked in racks that are placed in rows. A colocation center is a type of datacenter where equipment, space, and network bandwidth are available for rental to customers.

(4) A datacenter typically includes a cooling system to enable the computing devices within the

datacenter to continue working within their specified thermal limits. Many computing devices use air for cooling system components. This requires datacenters to utilize air-based cooling techniques to transfer the heat to the external environment. Air has a low specific heat capacity, which means that large amounts of air are required to remove a watt of heat. Air-based cooling techniques often require expensive infrastructure components such as computer room air conditioning (CRAC) units, air conditioner compressors, air circulation fans, duct work, air handlers, dehumidifiers, and the like.

(5) One of the challenges in managing a datacenter is achieving the right balance between space and power. For various reasons, power density has increased considerably over the last few years. As server racks become more thermally dense requiring greater quantities of air for cooling, the costs and logistics of conventional cooling using air becomes increasingly challenging.

(6) Some datacenters utilize immersion cooling techniques in which computing devices are submerged in a thermally conductive, electrically isolating dielectric fluid, which may be referred to as a dielectric working fluid or a heat transfer fluid. In an immersion cooling system, at least one container (e.g., a tank) is filled with the dielectric working fluid, and computing devices are placed in the container. The container may be referred to herein as an immersion tank. The dielectric working fluid has a higher heat capacity than air. Thus, the dielectric working fluid requires less fluid volume for a given heat load than air does.

(7) Heat can be removed from the heat-generating components (e.g., integrated circuits) within the computing devices by circulating the dielectric working fluid into direct contact with the heat-generating components, then through heat exchangers where the waste heat is transferred. Fluids suitable for immersion cooling have very good insulating properties to ensure that they can safely come into contact with energized electronic components without significantly altering the electrical characteristics of the system or system components. Immersion cooling has the potential to become a popular cooling solution for datacenters because it allows operators to drastically reduce their energy usage through the elimination of the air cooling infrastructure.

(8) Broadly speaking, there are two different types of immersion cooling techniques: single-phase immersion cooling and two-phase immersion cooling.

(9) With a single-phase immersion cooling system, the working fluid never changes state and always remains in a liquid form. In some implementations, the working fluid may be actively circulated by pumping the dielectric coolant in, through, and around the computing devices being cooled, and then transferring the heat absorbed by the coolant to a heat rejection device such as a radiator, dry cooler, liquid-to-liquid heat exchanger, or cooling tower. Alternatively, the working fluid may be passively circulated by the natural convection of the heated coolant to the heat rejection device(s).

(10) In a two-phase immersion cooling system, the heat of vaporization and the specific heat capacity characteristics of the dielectric working fluid are utilized for cooling. The dielectric working fluid generally has a relatively low boiling point such that heat absorbed by the dielectric working fluid surrounding the computing devices causes a portion of the dielectric working fluid to boil off or vaporize into a gas. The phase change of the dielectric working fluid carries heat away from the computing devices. The vapors produced by the boiling of the dielectric working fluid rise above the fluid pool where they contact a condenser that is cooler than the dielectric working fluid's boiling point. This causes the vapors to condense back into a liquid and fall back into the fluid pool.

SUMMARY

(11) In accordance with one aspect of the present disclosure, an immersion cooling system includes an immersion tank that is configured to retain dielectric working fluid and to hold a plurality of computing devices submerged in the dielectric working fluid. The immersion cooling system also includes a condenser that is configured to cause condensation of vaporized working fluid. The immersion cooling system also includes a subcooling heat exchanger that is in fluid communication

with a coolant source. The coolant source provides coolant having a coolant temperature that is lower than a boiling point of the dielectric working fluid. The subcooling heat exchanger is positioned so that heat transfer can occur between the dielectric working fluid and the subcooling heat exchanger. The immersion cooling system also includes a control system that controls how much of the coolant flows into the subcooling heat exchanger based at least in part on a temperature of the dielectric working fluid.

(12) In some embodiments, the condenser can be located in a vapor space of the immersion tank, and the subcooling heat exchanger can be submerged in the dielectric working fluid.

(13) In some embodiments, both the condenser and the subcooling heat exchanger can be embedded in an external wall of the immersion tank.

(14) In some embodiments, the control system can include a temperature sensor that measures the temperature of the dielectric working fluid, a valve that controls a flow of the coolant to the subcooling heat exchanger, and a controller that controls opening and closing of the valve based at least in part on the temperature of the dielectric working fluid.

(15) In some embodiments, the controller can calculate an error value as a difference between a desired temperature of the dielectric working fluid and the temperature of the dielectric working fluid as measured by the temperature sensor. The opening and the closing of the valve can be based at least in part on the error value.

(16) In some embodiments, the immersion cooling system additionally includes a pipe that is submerged in the dielectric working fluid. The pipe can include a plurality of nozzles and a pump that forces the dielectric working fluid to flow through the pipe. The pipe can be positioned to cause a plurality of streams of the dielectric working fluid to exit out of the plurality of nozzles in a direction of at least one heat-generating component on at least one computing device.

(17) In some embodiments, the control system can be additionally configured to proactively adjust coolant flow to the subcooling heat exchanger based at least in part on an expected workload of the plurality of computing devices.

(18) In some embodiments, the immersion tank can additionally include a diaphragm that can affect a vapor space pressure in the vapor space of the immersion tank. Increasing the vapor space pressure can cause a saturation temperature of the dielectric working fluid to increase. Decreasing the vapor space pressure can cause the saturation temperature of the dielectric working fluid to decrease.

(19) In some embodiments, the diaphragm can be located in the vapor space of the immersion tank. The condenser and the subcooling heat exchanger can be embedded in an external wall of the immersion tank.

(20) In some embodiments, the immersion cooling system can additionally include a relief valve connected to the diaphragm. The relief valve can be configured to release pressure from the diaphragm when the vapor space pressure equals the diaphragm pressure.

(21) In accordance with another aspect of the present disclosure, an immersion cooling system includes an immersion tank that is configured to retain dielectric working fluid and to hold a plurality of computing devices submerged in the dielectric working fluid. The immersion tank includes a vapor space above the dielectric working fluid. The immersion cooling system also includes a condenser that is configured to cause condensation of vaporized working fluid. The immersion cooling system also includes a subcooling heat exchanger that is in fluid communication with a coolant source. The coolant source provides coolant having a coolant temperature that is lower than a boiling point of the dielectric working fluid. The subcooling heat exchanger is positioned so that heat transfer can occur between the dielectric working fluid and the subcooling heat exchanger. The immersion cooling system also includes a diaphragm that changes a vapor space pressure in the vapor space of the immersion tank. Changing the vapor space pressure also changes a saturation temperature of the dielectric working fluid.

(22) In some embodiments, the diaphragm can be located in the vapor space of the immersion tank,

and the condenser and the subcooling heat exchanger can be embedded in an external wall of the immersion tank.

(23) In some embodiments, the immersion cooling system can additionally include a relief valve connected to the diaphragm. The relief valve can be configured to release pressure from the diaphragm when the vapor space pressure equals the diaphragm pressure.

(24) In some embodiments, the condenser can be located in the vapor space of the immersion tank, and the subcooling heat exchanger can be submerged in the dielectric working fluid.

(25) In some embodiments, the immersion cooling system can additionally include a control system that controls how much of the coolant flows into the subcooling heat exchanger based at least in part on a temperature of the dielectric working fluid.

(26) In some embodiments, the control system can include a temperature sensor that measures the temperature of the dielectric working fluid, a valve that controls a flow of the coolant to the subcooling heat exchanger, and a controller that controls opening and closing of the valve based at least in part on the temperature of the dielectric working fluid.

(27) In some embodiments, the controller can calculate an error value as a difference between a desired temperature of the dielectric working fluid and the temperature of the dielectric working fluid as measured by the temperature sensor. The opening and the closing of the valve can be based at least in part on the error value.

(28) In some embodiments, the immersion cooling system can additionally include a pipe that is submerged in the dielectric working fluid. The pipe can include a plurality of nozzles and a pump that forces the dielectric working fluid to flow through the pipe. The pipe can be positioned to cause a plurality of streams of the dielectric working fluid to exit out of the plurality of nozzles in a direction of at least one heat-generating component on at least one computing device.

(29) Another aspect of the present disclosure is directed to a method for subcooling dielectric working fluid in an immersion cooling system that includes a plurality of computing devices submerged in the dielectric working fluid. The method can include determining a desired temperature of the dielectric working fluid. The desired temperature of the dielectric working fluid can be less than a boiling point of the dielectric working fluid. The method can also include calculating an error value as a difference between the desired temperature of the dielectric working fluid and a measured temperature of the dielectric working fluid. The method can also include causing coolant to flow to a subcooling heat exchanger based at least in part on the error value when the measured temperature of the dielectric working fluid is greater than the desired temperature of the dielectric working fluid. The method can also include preventing the coolant from flowing to the subcooling heat exchanger when the measured temperature of the dielectric working fluid is less than or equal to the desired temperature of the dielectric working fluid.

(30) In some embodiments, the immersion cooling system can additionally include a diaphragm having a diaphragm pressure and a relief valve connected to the diaphragm. The method can additionally include pressurizing the diaphragm to increase pressure in a vapor space of the immersion tank. Increasing the pressure in the vapor space of the immersion tank can cause a saturation temperature of the dielectric working fluid to increase. The method can additionally include causing the relief valve to release pressure from the diaphragm when the pressure in the vapor space of the immersion tank equals the diaphragm pressure.

(31) In some embodiments, the method can include increasing a flow rate of the coolant to the subcooling heat exchanger in response to overclocking at least one component in at least one computing device of the plurality of computing devices.

(32) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(33) Additional features and advantages will be set forth in the description that follows. Features

and advantages of the disclosure may be realized and obtained by means of the systems and methods that are particularly pointed out in the appended claims. Features of the present disclosure will become more fully apparent from the following description and appended claims, or may be learned by the practice of the disclosed subject matter as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other features of the disclosure can be obtained, a more particular description will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. For better understanding, the like elements have been designated by like reference numbers throughout the various accompanying figures. Understanding that the drawings depict some example embodiments, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 illustrates an example of an immersion cooling system in accordance with at least one embodiment of the present disclosure.

(3) FIG. 2 illustrates an example of a method for subcooling dielectric working fluid in an immersion cooling system in accordance with at least one embodiment of the present disclosure.

(4) FIG. 3 illustrates another example of an immersion cooling system in accordance with at least one embodiment of the present disclosure.

(5) FIG. 4 illustrates an example of a method for using pressure manipulation to increase the heat flux at the heat-generating components of computing devices in an immersion cooling system in accordance with at least one embodiment of the present disclosure.

(6) FIG. 5A illustrates a perspective view of another example of an immersion cooling system in accordance with at least one embodiment of the present disclosure.

(7) FIG. 5B is a perspective view of the immersion cooling system shown in FIG. 5A with a lid shown on top of the immersion tank.

DETAILED DESCRIPTION

(8) The present disclosure is generally related to an improved immersion cooling system that can facilitate greater heat transfer away from heat-generating components of computing devices (e.g., integrated circuits such as CPUs) than is possible with currently available immersion cooling systems. More specifically, the techniques disclosed herein can enable greater heat flux at the heat-generating components of computing devices than is possible with currently available immersion cooling systems.

(9) An immersion cooling system in accordance with the present disclosure can utilize subcooling to increase the heat flux at the heat-generating components of computing devices. The term “subcooling” refers to a liquid existing at a temperature below its normal boiling point.

(10) When components of a computing device that is submerged in dielectric working fluid generate heat and cause the dielectric working fluid to boil, this causes vapor bubbles to be generated at or near the surface of the heat-generating components. Subcooling the dielectric working fluid can affect both the size of the vapor bubbles that are generated and the frequency at which they are generated. More specifically, subcooling the dielectric working fluid can increase the frequency of bubble generation. In addition, subcooling the dielectric working fluid can reduce the diameter of the vapor bubbles that are generated. This is at least in part because the subcooled conditions cause bubbles departing from the surface of a heat-generating component to shrink as they come in contact with cooler surrounding liquid. Both of these effects (increasing the frequency at which vapor bubbles are generated, and decreasing the diameter of the vapor bubbles that are generated) can increase the maximum heat load the dielectric working fluid can carry away from a

heat-generating component.

(11) Heat transfer through the boiling of a dielectric fluid, in the manner described above, is limited by a condition called the “critical heat flux.” In the context of immersion cooling, the critical heat flux can refer to a point where the amount of heat that is generated by a component of a computing device becomes sufficiently large that the departure rate of bubbles (formed from boiling of the dielectric working fluid) becomes insufficient to cool the component. Advantageously, subcooling can extend the critical heat flux of an immersion cooling system. In other words, subcooling can enable more heat to be transferred before the critical heat flux limitation occurs than would be the case in the absence of subcooling.

(12) One or more heat exchangers can be used to facilitate subcooling. As noted above, a two-phase immersion cooling system typically includes one or more heat exchangers in the form of condensers that are positioned to come into contact with vaporized working fluid, thereby causing the vaporized working fluid to condense back into a liquid and fall back into the pool of dielectric working fluid. In accordance with the present disclosure, however, an immersion cooling system can also include one or more additional heat exchangers for the purpose of facilitating subcooling. The heat exchanger(s) that are used to facilitate subcooling can be separate from the condensers that cause the working fluid vapors to condense back into the pool of dielectric fluid. A heat exchanger that facilitates subcooling may be referred to herein as a subcooling heat exchanger.

(13) An immersion cooling system in accordance with the present disclosure can include one or more subcooling heat exchangers that are positioned so that heat transfer can occur between the dielectric working fluid and the subcooling heat exchanger(s). In some embodiments, one or more subcooling heat exchangers can be submerged in the dielectric working fluid. In some embodiments, one or more subcooling heat exchangers can be embedded in an external wall of the immersion tank.

(14) A subcooling heat exchanger can be in fluid communication with a source of coolant. The coolant source can provide coolant having a temperature that is lower than a boiling point of the dielectric working fluid. The immersion cooling system can also include a control system that controls how much of the coolant flows into the subcooling heat exchanger based at least in part on the temperature of the dielectric working fluid. The control system can include a temperature sensor that measures the temperature of the dielectric working fluid, a valve that controls a flow of the coolant to the subcooling heat exchanger, and a controller that controls opening and closing of the valve based at least in part on the temperature of the dielectric working fluid.

(15) In some embodiments, the controller can be configured to calculate an error value as a difference between a desired temperature of the dielectric working fluid and the temperature of the dielectric working fluid as measured by the temperature sensor. The opening and the closing of the valve can be based at least in part on the error value. For example, if the desired temperature of the dielectric working fluid is greater than or equal to the measured temperature, then the controller can cause the valve to be closed in order to prevent coolant from flowing to the subcooling heat exchanger. On the other hand, if the measured temperature of the dielectric working fluid exceeds the desired temperature, then the controller can cause the valve to be opened so that coolant can flow to the subcooling heat exchanger.

(16) In some embodiments, an immersion cooling system in accordance with the present disclosure can also utilize the manipulation of pressure to increase the heat flux at the heat-generating components of computing devices. In other words, an immersion cooling system in accordance with the present disclosure can utilize a combination of subcooling and the manipulation of pressure to achieve the benefits described herein. As with subcooling, the manipulation of pressure can also extend the critical heat flux of an immersion cooling system.

(17) Pressure manipulation can be accomplished through the use of a diaphragm. In some embodiments, the diaphragm can be located in the vapor space of the immersion tank. In this context, the vapor space of the immersion tank can refer to the part of the immersion tank that is

directly above the pool of dielectric working fluid. When the dielectric working fluid boils, vapors can rise into the vapor space of the immersion tank.

(18) The diaphragm can be separated from the rest of the immersion tank. Having the diaphragm in the immersion tank can cause the pressure in the rest of the immersion tank to increase. This can cause the saturation temperature of the dielectric working fluid to increase. Advantageously, increasing the saturation temperature of the dielectric working fluid can increase the heat flux at the heat-generating components of the computing devices that are submerged in the dielectric working fluid.

(19) A relief valve can be connected to the diaphragm. The relief valve can be configured to release pressure from the diaphragm when the pressure in the vapor space of the immersion tank equals the pressure in the diaphragm.

(20) As noted above, the techniques disclosed herein can enable greater heat flux at the heat-generating components of computing devices than is possible with currently available immersion cooling systems. One of the benefits of this additional heat flux is that it allows the computing devices to be utilized to a greater extent than would otherwise be possible. For example, the additional heat flux provided by the techniques disclosed herein can enable the use of overclocking.

(21) In general terms, overclocking is the practice of increasing the clock rate of a computer to exceed that certified by the manufacturer. More specifically, overclocking increases the operating speed of a given component. Normally, overclocking is used to increase the performance of a major chip or subsystem, such as the main processor or graphics controller. However, overclocking can also be used in connection with other components, such as system memory and/or system buses.

(22) Overclocking causes an increase in power consumption. When a particular component is overclocked, higher current and voltage are applied to the component so that its power consumption increases. This can create thermal challenges.

(23) Components of a computing device are typically designed for normal operation, without overclocking. When a particular component is overclocked, that component is pushed farther from a power perspective, thereby causing additional heat to be generated. An overclocked component may be unreliable or fail completely if the additional heat load is not removed. The techniques disclosed herein, including subcooling and pressure manipulation, can be used to more effectively remove the additional heat that is produced by overclocking so that overclocking can be safely utilized.

(24) There are many benefits that can be achieved through the use of overclocking. As just one example, consider a scenario where customers rent the use of virtual machines (VMs) that run on servers maintained by a cloud computing provider. By overclocking the processors in the servers, more VMs could be created. Alternatively, the same number of VMs could be created, but the VMs could have improved performance. Alternatively still, some combination of more VMs and improved performance could be achieved. The techniques disclosed herein enable these benefits to be realized by facilitating the use of overclocking in a safe, sustainable way.

(25) FIG. 1 illustrates an example of an immersion cooling system **100** in accordance with at least one embodiment of the present disclosure. The immersion cooling system **100** includes an immersion tank **102**. The immersion tank **102** is configured to hold a plurality of computing devices. For simplicity, FIG. 1 shows a single computing device in the immersion tank **102**. In particular, FIG. 1 shows a motherboard **104** in the immersion tank **102**. A plurality of heat-generating components are located on the motherboard **104**. The heat-generating components can include integrated circuits **106** on the motherboard **104**.

(26) The immersion tank **102** is also configured to retain dielectric working fluid **108**. The computing devices are positioned in the immersion tank **102** such that the computing devices are submerged in the dielectric working fluid **108**. FIG. 1 shows the motherboard **104** submerged in the dielectric working fluid **108**.

(27) As discussed above, the computing devices in the immersion tank **102** generate heat during

normal operation. The dielectric working fluid **108** has a relatively low boiling point such that heat absorbed by the dielectric working fluid **108** surrounding the computing devices causes a portion of the dielectric working fluid **108** to boil off or vaporize into a gas. The vapors produced by the boiling of the dielectric working fluid **108** rise above the fluid pool into a part of the immersion tank **102** that may be referred to herein as a vapor space **110**. The vapor space **110** is located above the dielectric working fluid **108**. FIG. **1** shows vaporized working fluid **112** rising up above the pool of dielectric working fluid **108**. As the vaporized working fluid **112** continues to rise, it will enter the vapor space **110** of the immersion tank **102**.

(28) The immersion cooling system **100** includes a condenser **114** located in the vapor space **110** of the immersion tank **102**. The condenser **114** is in fluid communication with a coolant source **116**. The coolant source **116** can be any system that is capable of supplying coolant to the immersion cooling system **100**.

(29) In some embodiments, the coolant can be any fluid that has a higher thermal transfer rate than the dielectric working fluid **108**, and is a liquid at and below the boiling point of the dielectric working fluid **108**. One example of a coolant that could be used is water. However, this example should not be interpreted as limiting the scope of the present disclosure. There are many other fluids that could be used as the coolant.

(30) When the coolant source **116** supplies the coolant to the immersion cooling system **100**, the temperature of the coolant is lower than the boiling point of the dielectric working fluid **108**. Therefore, as the coolant flows through the condenser **114**, the condenser **114** is cooled. The condenser **114** is positioned so that it will cause condensation of vaporized working fluid **112**. In the depicted embodiment, the condenser **114** is positioned directly above the dielectric working fluid **108**, so that it will come into direct contact with vaporized working fluid **112**. Therefore, when coolant flows through the condenser **114**, the condenser **114** will cause the vaporized working fluid **112** to condense back into a liquid and fall back into the pool of dielectric working fluid **108**.

(31) A pump (not shown) causes the coolant to flow through the condenser **114**. For purposes of the present discussion, it will be assumed that the coolant flows from left to right in FIG. **1**. Coolant enters an inlet **118** of the condenser **114**. As vaporized working fluid **112** comes into contact with the condenser **114**, the coolant is heated. Heated coolant exits out of an outlet **120** of the condenser **114**. FIG. **1** shows the heated coolant being returned to a coolant return **122**. The coolant return **122** can be any device or system that is capable of receiving heated coolant from the immersion cooling system **100**.

(32) The arrows from the coolant source **116** to the inlet **118** of the condenser **114** represent one or more pipes that deliver the coolant from the coolant source **116** to the inlet **118** of the condenser **114**. Similarly, the arrows from the outlet **120** of the condenser **114** to the coolant return **122** represent one or more pipes that deliver the heated coolant from the outlet **120** of the condenser **114** to the coolant return **122**.

(33) In addition to the condenser **114**, the immersion cooling system **100** also includes another heat exchanger. This other heat exchanger may be referred to herein as a subcooling heat exchanger **124**. Whereas the condenser **114** causes condensation of vaporized working fluid **112**, the subcooling heat exchanger **124** is positioned so that it will facilitate subcooling of the dielectric working fluid **108**. In other words, the subcooling heat exchanger **124** is positioned so that it will cause heat transfer to occur between the dielectric working fluid **108** and the subcooling heat exchanger **124**. In the depicted embodiment, the subcooling heat exchanger **124** is submerged in the dielectric working fluid **108**, so that the subcooling heat exchanger **124** will come into direct contact with the dielectric working fluid **108**.

(34) Like the condenser **114**, the subcooling heat exchanger **124** is also in fluid communication with the coolant source **116**. A pump (not shown) causes the coolant to enter an inlet **126** of the subcooling heat exchanger **124** and flow through the subcooling heat exchanger **124**. As the coolant flows through the subcooling heat exchanger **124**, the subcooling heat exchanger **124** will subcool

the dielectric working fluid **108**. As discussed above, subcooling the dielectric working fluid **108** can increase the frequency of bubble generation while also reducing the diameter of the vapor bubbles that are generated. Both of these effects can increase the maximum heat load the dielectric working fluid **108** can carry away from the heat-generating components (e.g., the integrated circuits **106**) on the motherboard **104**. As the coolant flows through the subcooling heat exchanger **124** and it subcools the dielectric working fluid **108**, the coolant is heated. Heated coolant exits out of an outlet **128** of the subcooling heat exchanger **124**. FIG. 1 shows the heated coolant being returned to the coolant return **122**.

(35) In the depicted embodiment, the same coolant source **116** is shown providing coolant to the condenser **114** and also to the subcooling heat exchanger **124**. In alternative embodiments, however, different coolant sources could be used. For example, a first coolant source could provide coolant to the condenser **114**, and a second coolant source that is different from the first coolant source could provide coolant to the subcooling heat exchanger **124**.

(36) Similarly, in the depicted embodiment, the same coolant return **122** is shown receiving coolant from the condenser **114** and from the subcooling heat exchanger **124**. In alternative embodiments, however, different coolant returns could be used. For example, a first coolant return could receive coolant from the condenser **114**, and a second coolant return that is different from the first coolant return could receive coolant from the subcooling heat exchanger **124**.

(37) The immersion cooling system **100** includes a control system **130** that controls the extent to which the dielectric working fluid **108** is subcooled. More specifically, the control system **130** is configured to control how much of the coolant flows into the subcooling heat exchanger **124** based at least in part on the temperature of the dielectric working fluid **108**. The control system **130** includes a valve **132**, a controller **134**, and a temperature sensor **136**. The valve **132** controls the flow of coolant to the subcooling heat exchanger **124**. The controller **134** controls the opening and the closing of the valve **132** based at least in part on the temperature of the dielectric working fluid **108**. In FIG. 1, the controller **134** is shown connected to a power supply **150**. The temperature sensor **136** measures the temperature of the dielectric working fluid **108**.

(38) When the valve **132** is open, coolant can flow to the subcooling heat exchanger **124**. Conversely, when the valve **132** is closed, coolant is prevented from flowing to the subcooling heat exchanger **124**. In some embodiments, the valve **132** can be capable of one or more intermediate states between fully opened and fully closed. For example, the valve **132** can be capable of being partially opened. When the valve **132** is partially opened, the rate of coolant flow can be greater than zero but less than when the valve **132** is fully opened.

(39) In some embodiments, the controller **134** can be configured to continuously calculate an error value. The error value can be calculated as the difference between (a) the desired temperature of the dielectric working fluid **108**, and (b) the actual temperature of the dielectric working fluid **108** (e.g., as measured by the temperature sensor **136**). To facilitate subcooling, the desired temperature of the dielectric working fluid **108** can be set to be below the boiling point of the dielectric working fluid **108**.

(40) The controller **134** can then open or close the valve **132** depending on the error value that is calculated. For example, if (a) is greater than or equal to (b), then the controller **134** can close the valve **132**. On the other hand, if (b) exceeds (a), then the controller **134** can open the valve **132** and allow coolant to flow to the subcooling heat exchanger **124**. In some embodiments, the extent to which the valve **132** is opened can depend at least in part on the extent to which (b) exceeds (a). For example, if (b) greatly exceeds (a), then the valve **132** can be opened to a greater extent than if (b) only slightly exceeds (a).

(41) In some embodiments, the controller **134** can be implemented as a proportional-integral-derivative (PID) controller. A PID controller continuously calculates an error value (such as the error value described above) and applies a correction based on proportional, integral, and derivative terms.

(42) In the depicted embodiment, the valve **132** controls the supply of coolant to the subcooling heat exchanger **124**, but does not control the supply of coolant to the condenser **114**. In some alternative embodiments, a valve could be provided that controls the supply of coolant to both the subcooling heat exchanger **124** and also to the condenser **114**. In some other alternative embodiments, at least two different valves could be provided. A first valve could control the supply of coolant to the subcooling heat exchanger **124**, and a second valve that is different from the first valve could control the supply of coolant to the condenser **114**.

(43) In some embodiments, the control system **130** can be configured to use information about the expected or incoming workload of the computing devices in the immersion tank **102** for the purpose of controlling the flow of coolant to the subcooling heat exchanger **124**. In other words, information about the expected or incoming workload can be used proactively to control the flow of coolant to the subcooling heat exchanger **124**.

(44) As noted above, the immersion cooling system **100** can be used in a datacenter. The datacenter can include a plurality of immersion tanks like the immersion tank **102** shown in FIG. **1**. A workload allocator **160** can be responsible for allocating workloads among the computing devices in the various immersion tanks. For example, the workload allocator **160** can be responsible for allocating virtual machines (VMs) among the computing devices in the various immersion tanks. The control system **130** can be in electronic communication with the workload allocator **160**. The workload allocator **160** can include a predictive element **162** that provides for proactive subcooling. The workload allocator **160** can provide information and/or signals to the control system **130** that cause the flow of coolant to the subcooling heat exchanger **124** to be adjusted based on the expected or predicted workload to be experienced by the computing devices in the immersion tank **102**.

(45) In one possible scenario, the predictive element **162** of the workload allocator **160** could predict that the computing devices in the immersion tank **102** will have increased workload (e.g., that additional VMs will be allocated) at a certain time of day. Based at least in part on this prediction, the workload allocator **160** can provide information and/or signals to the control system **130** that cause the flow of coolant to the subcooling heat exchanger **124** to be increased during this time of day.

(46) The opposite scenario could also occur. The predictive element **162** of the workload allocator **160** could predict that the computing devices in the immersion tank **102** will have decreased workload (e.g., that fewer VMs will be allocated) at a certain time of day. Based at least in part on this prediction, the workload allocator **160** can provide information and/or signals to the control system **130** that cause the flow of coolant to the subcooling heat exchanger **124** to be decreased during this time of day.

(47) Predictions made by the predictive element **162** of the workload allocator **160** can be based at least in part on historical data about the workload experienced by the computing devices in the immersion tank **102** (and in other immersion tanks in the immersion cooling system **100**). In some embodiments, machine learning techniques can be utilized to make such predictions.

(48) As discussed above, the additional heat flux provided by the techniques disclosed herein can enable the use of overclocking. In some embodiments, the predictive element **162** of the workload allocator **160** can predict when overclocking will be utilized. When a prediction is made that overclocking will be utilized during a particular period of time, the workload allocator **160** can provide information and/or signals to the control system **130** that cause the flow of coolant to the subcooling heat exchanger **124** to be increased to compensate for the additional power consumption caused by overclocking.

(49) The immersion cooling system **100** shown in FIG. **1** includes just one subcooling heat exchanger **124** submerged in the dielectric working fluid **108**. In some alternative embodiments, an immersion cooling system **100** could include a plurality of subcooling heat exchangers submerged in dielectric working fluid **108**. In some embodiments, the plurality of subcooling heat exchangers

could be in fluid communication with the same coolant source and the same coolant return. In other embodiments, the plurality of subcooling heat exchangers could be in fluid communication with different coolant sources and/or with different coolant returns.

(50) The immersion cooling system **100** also includes a pump **138** that circulates the dielectric working fluid **108** throughout the immersion tank **102**. Circulating the dielectric working fluid **108** throughout the immersion tank **102** helps to ensure that the temperature of the dielectric working fluid **108** is substantially uniform.

(51) In addition to circulating the dielectric working fluid **108** throughout the immersion tank **102**, the pump **138** also causes the dielectric working fluid **108** to flow through a pipe **140** that is positioned near the computing devices that are being cooled. In FIG. **1**, where only a single computing device (i.e., the motherboard **104**) is shown, the pipe **140** is positioned close to that computing device. The pipe **140** includes a plurality of nozzles **142**. When the pump **138** causes the dielectric working fluid **108** to flow through the pipe **140**, this causes streams of dielectric working fluid **108** to exit out of the nozzles **142** in the direction of the heat-generating components (e.g., the integrated circuits **106**) on the motherboard **104**. In addition to helping to ensure that the temperature of the dielectric working fluid **108** is substantially uniform, aiming streams of dielectric working fluid **108** at the heat-generating components on the motherboard **104** also helps to cool the heat-generating components.

(52) The embodiment shown in FIG. **1** includes just a single pump **138** and a single pipe **140** with nozzles **142**. However, an immersion cooling system **100** in accordance with the present disclosure could include a plurality of pumps and/or a plurality of pipes with nozzles. The pumps and/or the pipes with nozzles could be placed in a variety of different locations throughout the immersion tank **102**.

(53) In FIG. **1**, the pipe **140** with the nozzles **142** is located beneath the bottom side of the motherboard **104**. In an alternative embodiment, a pipe with nozzles could be located near the left side, the right side, and/or the top side of the motherboard **104**. If the immersion tank **102** includes a plurality of computing devices, pipes with nozzles can be located in similar positions with respect to the plurality of computing devices.

(54) In FIG. **1**, the pump **138** is located near an inlet **144** to the pipe **140** with the nozzles **142**. In embodiments that include a plurality of pipes similar to the pipe **140** shown in FIG. **1**, a plurality of pumps could be provided. The pumps could be positioned such that at least one pump is located near at least one inlet to each pipe in the immersion tank **102**.

(55) The immersion cooling system **100** includes a temperature sensor **146** that is configured to measure the temperature of the vapor space **110** of the immersion tank **102**. The immersion cooling system **100** also includes a pressure sensor **148** that is configured to measure the pressure of the vapor space **110** of the immersion tank **102**. The temperature sensor **146** and the pressure sensor **148** can be used to control the rate of coolant flow through the condenser **114**. More specifically, based on the temperature that is measured, the temperature sensor **146** can send a signal to a controller (not shown) that controls the rate of coolant flow through the condenser. Similarly, based on the pressure that is measured, the pressure sensor **148** can also send a signal to the controller. The controller can control the rate of coolant flow through the condenser **114** based at least in part on the signals received from the temperature sensor **146** and the pressure sensor **148**.

(56) FIG. **2** illustrates an example of a method **200** for subcooling dielectric working fluid in an immersion cooling system in accordance with at least one embodiment of the present disclosure. For the sake of clarity, the method **200** will be described in relation to the immersion cooling system **100** shown in FIG. **1**. The method **200** can be implemented by a control system, such as the control system **130** in the immersion cooling system **100** of FIG. **1**.

(57) The method **200** can include determining **201** the desired temperature of the dielectric working fluid **108** ($T_{sub.desired}$). In some embodiments, $T_{sub.desired}$ can be set to be a pre-determined amount below the saturation temperature ($T_{sub.sat}$) of the dielectric working fluid **108**. For

example, $T_{sub.desired}$ can be set to be N degrees below $T_{sub.sat}$, where N can be a configurable parameter. In some embodiments, N can be at least 10°C . In other words, in some embodiments $T_{sub.desired}$ can be set to be at least 10°C . below $T_{sub.sat}$.

(58) The value of $T_{sub.sat}$ varies based on the type of dielectric working fluid **108** that is being used. As an example, a typical range for $T_{sub.sat}$ is from about 34°C . to about 60°C .

(59) The method **200** can also include determining **203** the measured temperature of the dielectric working fluid **108** ($T_{sub.measured}$). In some embodiments, determining $T_{sub.measured}$ can include receiving one or more temperature measurements from one or more temperature sensors, such as the temperature sensor **136** shown in FIG. **1**.

(60) The method **200** can also include calculating **205** an error value as the difference between $T_{sub.desired}$ and $T_{sub.measured}$. One skilled in the art will understand that calculating the difference between $T_{sub.desired}$ and $T_{sub.measured}$ can be achieved by performing the calculation $T_{sub.desired}-T_{sub.measured}$ or by performing the calculation $T_{sub.measured}-T_{sub.desired}$.

(61) The method **200** can also include determining **207** how to adjust the flow of coolant to the subcooling heat exchanger **124** based at least in part on the error value that is calculated. For example, when $T_{sub.measured}$ is greater than $T_{sub.desired}$, the method **200** can include causing **209** the coolant to flow to the subcooling heat exchanger **124** based at least in part on the error value that is calculated. On the other hand, when $T_{sub.desired}$ is greater than or equal to $T_{sub.measured}$, the method **200** can include preventing **211** the flow of coolant to the subcooling heat exchanger **124**.

(62) When $T_{sub.measured}$ is greater than $T_{sub.desired}$ and the method **200** includes causing **209** the coolant to flow to the subcooling heat exchanger **124** based at least in part on the error value that is calculated, there are several different ways that the rate of coolant flow can be determined. In some embodiments, the flow of coolant to the subcooling heat exchanger **124** can depend at least in part on the magnitude of the difference between $T_{sub.measured}$ and $T_{sub.desired}$. For example, the controller **134** can cause more coolant to flow to the subcooling heat exchanger **124** (or, in other words, cause coolant to flow to the subcooling heat exchanger **124** at a faster rate) when $T_{sub.measured}$ is significantly greater than $T_{sub.desired}$ as compared to when $T_{sub.measured}$ is only slightly greater than $T_{sub.desired}$.

(63) In some embodiments, the flow of coolant to the subcooling heat exchanger **124** can be directly proportional to the difference between $T_{sub.measured}$ and $T_{sub.desired}$. For example, consider a scenario in which $T_{sub.measured}-T_{sub.desired}=2n$ at time $t_{sub.1}$, and $T_{sub.measured}-T_{sub.desired}=n$ at time $t_{sub.2}$. In this scenario, the rate of coolant flow to the subcooling heat exchanger **124** could be twice as great at time $t_{sub.1}$ as it is at time $t_{sub.2}$.

(64) In some embodiments, causing **209** the coolant to flow to the subcooling heat exchanger **124** can include causing a valve **132** that controls the flow of the coolant to the subcooling heat exchanger **124** to be opened. The extent to which the valve **132** is opened can depend on the desired rate of coolant flow. For example, the valve **132** can be opened to a greater extent when a higher rate of coolant flow is desired as compared to when a lesser rate of coolant flow is desired.

(65) As discussed above, the additional heat flux provided by the techniques disclosed herein can enable the use of overclocking. When one or more computing devices in an immersion tank are overclocked, this can cause an increase in power consumption. The increase in power consumption can cause the temperature of the dielectric working fluid to increase. For example, reference is once again made to the immersion cooling system **100** shown in FIG. **1** and the method **200** shown in FIG. **2**. If one or more components (e.g., a CPU) on the motherboard **104** is overclocked, the resulting increase in power consumption can cause the temperature of the dielectric working fluid **108** in the immersion tank **102** to increase. The increase in power consumption can cause the error value (i.e., the difference between $T_{sub.desired}$ and $T_{sub.measured}$ that is calculated **205** in the method **200** shown in FIG. **2**) to increase. The increase in the error value can therefore cause **209**

the coolant to flow to the subcooling heat exchanger **124** at a higher rate.

(66) FIG. **3** illustrates another example of an immersion cooling system **300** in accordance with at least one embodiment of the present disclosure. The immersion cooling system **300** is similar in several respects to the immersion cooling system **100** shown in FIG. **1**. For example, the immersion cooling system **300** includes an immersion tank **302** that is configured to hold a plurality of computing devices. For the sake of simplicity, FIG. **3** shows a single computing device in the immersion tank **302**, namely a motherboard **304**. A plurality of heat-generating components (e.g., integrated circuits **306**) are located on the motherboard **304**. The immersion tank **302** is also configured to retain dielectric working fluid **308** such that the computing devices in the immersion tank **302** are submerged in the dielectric working fluid **308**. FIG. **3** shows the motherboard **304** submerged in the dielectric working fluid **308**.

(67) The immersion cooling system **300** also includes a pump **338**. In addition to circulating the dielectric working fluid **308** throughout the immersion tank **302**, the pump **338** also causes the dielectric working fluid **308** to flow through a pipe **340** that is positioned near the computing devices that are being cooled. The pipe **340** includes a plurality of nozzles **342**. When the pump **338** causes the dielectric working fluid **308** to flow through the pipe **340**, this causes streams of dielectric working fluid **308** to exit out of the nozzles **342** in the direction of the heat-generating components (e.g., the integrated circuits **306**) on the motherboard **304**.

(68) The immersion cooling system **300** is shown with a temperature sensor **336** that is configured to measure the temperature of the dielectric working fluid **308**. The immersion cooling system **300** is also shown with a temperature sensor **346** that is configured to measure the temperature of the vapor space **310** of the immersion tank **302**.

(69) The immersion cooling system **300** is also different in many respects from the immersion cooling system **100** shown in FIG. **1**. For example, as discussed above, the immersion cooling system **100** of FIG. **1** includes a condenser **114** in the vapor space **110** of the immersion tank **102** and a subcooling heat exchanger **124** submerged in the dielectric working fluid **108**. In contrast, the immersion cooling system **300** shown in FIG. **3** does not include any heat exchangers within the internal part of the immersion tank **302**. Instead, the immersion cooling system **300** includes a plurality of condensers and a plurality of subcooling heat exchangers embedded in the external walls of the immersion tank **302**.

(70) More specifically, a first external wall **352-1** of the immersion tank **302** includes a plurality of channels that carry coolant from a coolant source **316**. A first plurality of upper channels **314-1** are positioned in an upper half of the immersion tank **302**, near the vapor space **310** of the immersion tank **302**. Because the first plurality of upper channels **314-1** are positioned near the vapor space **310** of the immersion tank **302**, the first plurality of upper channels **314-1** act as condensers. Thus, the first plurality of upper channels **314-1** cause condensation of vaporized working fluid **312**. A first plurality of lower channels **324-1** are positioned in a lower half of the immersion tank **302**, near the computing devices submerged in the dielectric working fluid **308**. Because the first plurality of lower channels **324-1** are positioned near the computing devices submerged in the dielectric working fluid **308**, the first plurality of lower channels **324-1** act as subcooling heat exchangers. Thus, the first plurality of lower channels **324-1** facilitate subcooling of the dielectric working fluid **308**.

(71) Similarly, a second external wall **352-2** of the immersion tank **302** includes a plurality of channels that carry heated coolant to a coolant return **322**. A second plurality of upper channels **314-2** are positioned in an upper half of the immersion tank **302**, near the vapor space **310** of the immersion tank **302**. A second plurality of lower channels **324-2** are positioned in a lower half of the immersion tank **302**, near the computing devices submerged in the dielectric working fluid **308**.

(72) Having the condensers (i.e., the first plurality of upper channels **314-1** and the second plurality of upper channels **314-2**) and the subcooling heat exchangers (i.e., the first plurality of lower channels **324-1** and the second plurality of lower channels **324-2**) outside of the immersion tank

302 creates additional space inside the immersion tank **302**. This additional space can be utilized for pressure manipulation techniques that facilitate greater heat transfer away from heat-generating components of computing devices.

(73) In the depicted embodiment, a diaphragm **354** can be included in the vapor space **310** of the immersion tank **302**. The diaphragm **354** can be formed in the lid **356** of the immersion tank **302**. A partition **358** can extend downward from the lid **356** and separate the diaphragm **354** from the rest of the immersion tank **302**.

(74) Because the diaphragm **354** is separated from the rest of the immersion tank **302**, the pressure inside the diaphragm **354** can be different from the pressure in the rest of the immersion tank **302**. In the discussion that follows, the pressure inside the diaphragm **354** may be referred to as $P_{\text{sub.dia}}$. The pressure in the vapor space **310** of the immersion tank **302** may be referred to as $P_{\text{sub.in/vap}}$.

(75) The diaphragm **345** can be used to manipulate $P_{\text{sub.in/vap}}$. In some situations, having the diaphragm **354** in the immersion tank **302** can cause $P_{\text{sub.in/vap}}$ to increase. In other words, $P_{\text{sub.in/vap}}$ can be higher with the diaphragm **354** in the immersion tank **302** than it would otherwise be without the diaphragm **354**. Moreover, increasing $P_{\text{sub.in/vap}}$ can cause the saturation temperature of the dielectric working fluid **308** ($T_{\text{sub.sat}}$) to increase. In this context, $T_{\text{sub.sat}}$ is the temperature for a corresponding saturation pressure ($P_{\text{sub.sat}}$) at which the dielectric working fluid **308** boils into its vapor phase. Advantageously, increasing $T_{\text{sub.sat}}$ of the dielectric working fluid **308** can increase the heat flux at the heat-generating components of the computing devices that are submerged in the dielectric working fluid **308**.

(76) In other situations, having the diaphragm **354** in the immersion tank **302** can cause $P_{\text{sub.in/vap}}$ to decrease. In other words, $P_{\text{sub.in/vap}}$ can be lower with the diaphragm **354** in the immersion tank **302** than it would otherwise be without the diaphragm **354**. Moreover, decreasing $P_{\text{sub.in/vap}}$ can cause the saturation temperature of the dielectric working fluid **308** ($T_{\text{sub.sat}}$) to decrease.

(77) Therefore, the presence of the diaphragm **354** in the immersion tank **302** can cause $P_{\text{sub.in/vap}}$ to increase or decrease, which can cause $T_{\text{sub.sat}}$ of the dielectric working fluid **308** to increase or decrease. Using the diaphragm **345** to manipulate $P_{\text{sub.in/vap}}$ in this way can have the effect of increasing the amount of heat that can be transferred by the immersion cooling system **300**.

(78) A relief valve **360** can be connected to the diaphragm **354**. The relief valve **360** can be configured to release pressure from the diaphragm **354** when $P_{\text{sub.dia}}$ equals $P_{\text{sub.in/vap}}$. More specifically, when $P_{\text{sub.in/vap}} < P_{\text{sub.dia}}$, pressure can continue to accumulate in the vapor space **310** of the immersion tank **302** such that $P_{\text{sub.sat}}$ of the dielectric working fluid **308** increases. When $P_{\text{sub.in/vap}}$ becomes equal to $P_{\text{sub.dia}}$, the relief valve **360** can release pressure from the diaphragm **354**.

(79) The immersion cooling system **300** includes a pressure sensor **362** that is configured to measure $P_{\text{sub.dia}}$ and a pressure sensor **348** that is configured to measure $P_{\text{sub.in/vap}}$.

(80) In the depicted embodiment, the diaphragm **354** is shown in the immersion tank **302**. However, in an alternative embodiment, the diaphragm could be located outside of the immersion tank. As long as one side of the diaphragm is connected to the headspace of the immersion tank (either directly or through a connecting orifice), the pressure of the headspace can be regulated without directly injecting additional material into the immersion tank. In some embodiments, a diaphragm that is located outside the immersion tank could be constructed through the use of a hydraulic system that includes an accumulator. In such embodiments, the pressure on the opposing side of the diaphragm could be regulated to change relative pressure and thus influence headspace pressure.

(81) FIG. 4 illustrates an example of a method **400** for using pressure manipulation to increase the heat flux at the heat-generating components of computing devices in an immersion cooling system

in accordance with at least one embodiment of the present disclosure. For the sake of clarity, the method **400** will be described in relation to the immersion cooling system **300** shown in FIG. 3.

(82) The method **400** can include pressurizing **401** the diaphragm **354**. The diaphragm **354** can be pressurized **401** to a point where the presence of the diaphragm **354** in the immersion tank **302** causes $P_{sub.in/vap}$ to be higher than it otherwise would be without the diaphragm **354**. As discussed above, causing $P_{sub.in/vap}$ to increase can cause $T_{sub.sat}$ of the dielectric working fluid **308** to increase, which can thereby increase the amount of heat that can be transferred by the immersion cooling system **300**.

(83) The method **400** can also include monitoring **403** the pressure inside the diaphragm **354** ($P_{sub.dia}$) and the pressure in the vapor space **310** of the immersion tank **302** ($P_{sub.in/vap}$). In some embodiments, monitoring **403** $P_{sub.dia}$ and $P_{sub.in/vap}$ can include receiving one or more pressure measurements from one or more pressure sensors.

(84) The method **400** can also include comparing $P_{sub.dia}$ and $P_{sub.in/vap}$ to determine **405** whether $P_{sub.dia}$ equals $P_{sub.in/vap}$. As long as $P_{sub.in/vap} < P_{sub.dia}$, pressure can continue to accumulate in the vapor space **310** of the immersion tank **302**. However, when it is determined **405** that $P_{sub.in/vap}$ is equal to $P_{sub.dia}$, the method **400** can include causing **407** the relief valve **360** to release pressure from the diaphragm **354**.

(85) FIG. 5A illustrates a perspective view of another example of an immersion cooling system **500** in accordance with at least one embodiment of the present disclosure. The immersion cooling system **500** represents one possible implementation of the immersion cooling system **300** shown in FIG. 3.

(86) The immersion cooling system **500** includes an immersion tank **502** that is configured to hold a plurality of computing devices **504**. The immersion tank **502** is configured to retain dielectric working fluid (not shown in FIGS. 5A-B) such that the computing devices in the immersion tank **502** are submerged in the dielectric working fluid.

(87) A first external wall **552-1** of the immersion tank **502** includes a plurality of channels that carry coolant from a coolant source (not shown in FIGS. 5A-B). A first plurality of upper channels **514-1** are positioned in an upper half of the immersion tank **502**, near the vapor space of the immersion tank **502**. The first plurality of upper channels **514-1** act as condensers that cause condensation of vaporized working fluid. A first plurality of lower channels **524-1** are positioned in a lower half of the immersion tank **502**, near the computing devices **504** submerged in the dielectric working fluid. The first plurality of lower channels **524-1** act as subcooling heat exchangers that facilitate subcooling of the dielectric working fluid.

(88) Similarly, a second external wall **552-2** of the immersion tank **502** includes a plurality of channels that carry heated coolant to a coolant return (not shown in FIGS. 5A-B). A second plurality of upper channels **514-2** are positioned in an upper half of the immersion tank **502**, near the vapor space of the immersion tank **502**. A second plurality of lower channels **524-2** are positioned in a lower half of the immersion tank **502**, near the computing devices **504** submerged in the dielectric working fluid.

(89) A pump (not shown) can cause the coolant to flow into the first plurality of upper channels **514-1** and the first plurality of lower channels **524-1**. The coolant in the first plurality of upper channels **514-1** causes the vaporized working fluid to condense, and the coolant in the first plurality of lower channels **524-1** subcools the dielectric working fluid. This causes the coolant to become heated. The second plurality of upper channels **514-2** and the second plurality of lower channels **524-2** carries heated coolant to a coolant return.

(90) FIG. 5B is a perspective view of the immersion cooling system **500** shown in FIG. 5A with a lid **556** shown on top of the immersion tank **502**. In some embodiments, a diaphragm (not shown in FIGS. 5A-B) can be formed in the lid **556**. The diaphragm can be similar to the diaphragm **354** shown in FIG. 3. As discussed above, the diaphragm can be used for pressure manipulation techniques that increase the heat flux at the heat-generating components of the computing devices

504 that are submerged in the dielectric working fluid.

(91) Some examples of various terms and phrases that have been used in the above discussion will now be provided.

(92) The term “heat flux” refers to the rate of thermal energy flow per unit surface area of a heat transfer surface.

(93) The term “immersion tank” may refer to any container that is sufficiently large and sturdy to retain a plurality of computing devices that are submerged in dielectric working fluid.

(94) The term “dielectric working fluid” (or simply “working fluid”) may refer to any nonconductive fluid in which computing devices can be submerged for the purpose of cooling the computing devices. Some examples of dielectric working fluids that can be used include synthetic fluids, fluorocarbon-based fluids, mineral oil, and deionized water. A dielectric working fluid may have a relatively low boiling point (e.g., 40-50° C.), such that heat generated by computing devices would normally cause the dielectric working fluid to boil.

(95) The term “condenser” may refer to any apparatus that may be used to condense vaporized working fluid. In some embodiments, a condenser may include one or more tubes, which may be shaped in the form of one or more coils. Cool liquid may be pumped through the tubes to facilitate condensation of vaporized working fluid.

(96) In some embodiments, two structures are in “fluid communication” with one another if there is a path for a fluid to flow between the two structures. In this context, the term “fluid” may refer generally to any substance that has the tendency to assume the shape of its container, and may include a liquid or a gas.

(97) The term “controller” may refer to hardware, software, firmware, or any combination thereof, that performs some or all of the steps, operations, actions, or other functionality disclosed herein in connection with the controller. If implemented in software, the techniques may be realized at least in part by a non-transitory computer-readable medium having computer-executable instructions stored thereon that, when executed by at least one processor, implement the disclosed functionality.

(98) As noted above, the “vapor space” of the immersion tank can refer to the part of the immersion tank that is directly above the pool of dielectric working fluid. In some contexts, the term “headspace” can be synonymous with the term “vapor space.” In other contexts, the term “headspace” can refer to a part of the immersion tank that is distinct from the vapor space. For example, the term “headspace” can refer to a part of the immersion tank that is above the vapor space.

(99) The steps, operations, and/or actions of the methods described herein may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps, operations, and/or actions is required for proper functioning of the method that is being described, the order and/or use of specific steps, operations, and/or actions may be modified without departing from the scope of the claims.

(100) The term “determining” (and grammatical variants thereof) can encompass a wide variety of actions. For example, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

(101) The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there can be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. For example, any element or feature described in relation to an embodiment herein may be combinable with any element or feature of any other embodiment described herein, where compatible.

(102) The present disclosure may be embodied in other specific forms without departing from its spirit or characteristics. The described embodiments are to be considered as illustrative and not restrictive. The scope of the disclosure is, therefore, indicated by the appended claims rather than by the foregoing description. Changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method for subcooling a dielectric working fluid in an immersion cooling system that comprises a plurality of means for computing, the method comprising: determining a desired temperature of the dielectric working fluid, wherein the desired temperature of the dielectric working fluid is less than a boiling point of the dielectric working fluid; calculating an error value as a difference between the desired temperature of the dielectric working fluid and a measured temperature of the dielectric working fluid; causing coolant to flow to a subcooling heat exchanger based at least in part on the error value when the measured temperature of the dielectric working fluid is greater than the desired temperature of the dielectric working fluid; and preventing the coolant from flowing to the subcooling heat exchanger when the measured temperature of the dielectric working fluid is less than or equal to the desired temperature of the dielectric working fluid.
2. The method of claim 1, further comprising increasing a flow rate of the coolant to the subcooling heat exchanger in response to overclocking at least one component of a means for generating heat in at least one means for computing of the plurality of means for computing.
3. The method of claim 1, further comprising adjusting a flow rate of the coolant to the subcooling heat exchanger in response to an expected workload of the plurality of means for computing.
4. The method of claim 1, wherein the subcooling heat exchanger is submerged in the dielectric working fluid.
5. The method of claim 1, wherein the subcooling heat exchanger is embedded in an external wall of an immersion tank of the immersion cooling system.
6. A method of transferring heat from a heat-generating component at least partially submerged in a dielectric working fluid in an immersion cooling system, the method comprising: pressurizing a diaphragm, wherein a means for generating heat and the dielectric working fluid are positioned in an immersion tank of the immersion cooling system, and wherein the diaphragm is positioned in a vapor space of the immersion tank; increasing a vapor space pressure of the vapor space; based on increasing the vapor space pressure, causing an increase in a saturation temperature of the dielectric working fluid; and causing an increase in a heat flux from the means for generating heat to the dielectric working fluid based on increasing the saturation temperature.
7. The method of claim 6, wherein increasing the heat flux by pressurizing the diaphragm is in response to overclocking the means for generating heat.
8. The method of claim 7, further comprising monitoring a diaphragm pressure of the diaphragm and monitoring the tank pressure.
9. The method of claim 8, further comprising continuing to pressurize the diaphragm based on determining that the diaphragm pressure is greater than the tank pressure.
10. The method of claim 8, further comprising releasing pressure from the diaphragm based on determining that the tank pressure is equal to the diaphragm pressure.
11. The method of claim 10, wherein releasing pressure from the diaphragm includes opening a pressure relief valve of the diaphragm.
12. The method of claim 6, wherein the immersion cooling system is a two-phase immersion cooling system.
13. An immersion cooling system, comprising: an immersion tank that is configured to retain a dielectric working fluid and to hold a plurality of means for computing submerged in the dielectric

working fluid; a condenser that is configured to cause condensation of vaporized working fluid; and a diaphragm that changes a vapor space pressure in a vapor space of the immersion tank, wherein changing the vapor space pressure also changes a saturation temperature of the dielectric working fluid.

14. The immersion cooling system of claim 13, further comprising a subcooling heat exchanger that is in fluid communication with a coolant source, the coolant source providing coolant having a coolant temperature that is lower than a boiling point of the dielectric working fluid, the subcooling heat exchanger being positioned so that heat transfer can occur between the dielectric working fluid and the subcooling heat exchanger.

15. The immersion cooling system of claim 14, further comprising a means for controlling how much of a flow of the coolant flows into the subcooling heat exchanger, wherein the flow of the coolant into the subcooling heat exchanger is based at least in part on a temperature of the dielectric working fluid.

16. The immersion cooling system of claim 14, wherein: the condenser is located in a vapor space of the immersion tank; and the subcooling heat exchanger is submerged in the dielectric working fluid.

17. The immersion cooling system of claim 14, wherein: the diaphragm is located in the vapor space of the immersion tank; and the condenser and the subcooling heat exchanger are embedded in an external wall of the immersion tank.

18. The immersion cooling system of claim 13, further comprising a relief valve for releasing a pressure of the diaphragm.

19. The immersion cooling system of claim 18, wherein the relief valve is configured to release the pressure of the diaphragm when the vapor space pressure equals a diaphragm pressure.

20. The immersion cooling system of claim 13, wherein: the immersion cooling system further comprises a pipe that is submerged in the dielectric working fluid, the pipe comprising a plurality of nozzles and a pump that forces the dielectric working fluid to flow through the pipe; and the pipe is positioned to cause a plurality of streams of the dielectric working fluid to exit out of the plurality of nozzles in a direction of at least one means for generating heat on at least one means for computing.
